SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Data Warehousing and Data Mining COURSE CODE: CSA1606

CO1-AT2 -Concept Map

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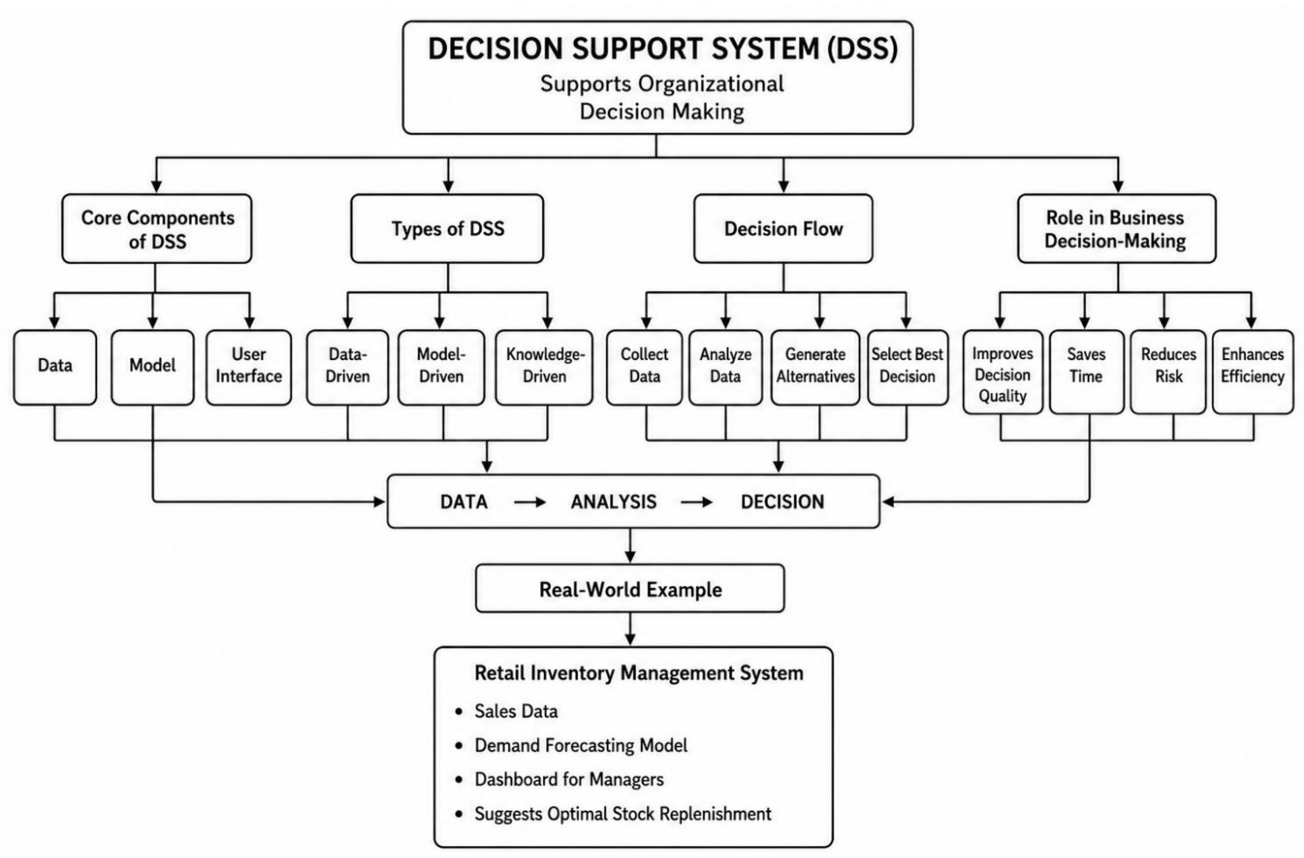
1) Decision Support Systems (DSS)

Task Description:
Construct a concept map
that explains how a Decision Support System (DSS) supports
organizational decision-making.

What your map must show:

1. Core components: Data, Model, User Interface
2. Types of DSS
3. Flow: Data → Analysis → Decision
4. Role in business decision-making
5. Real-world example linkage



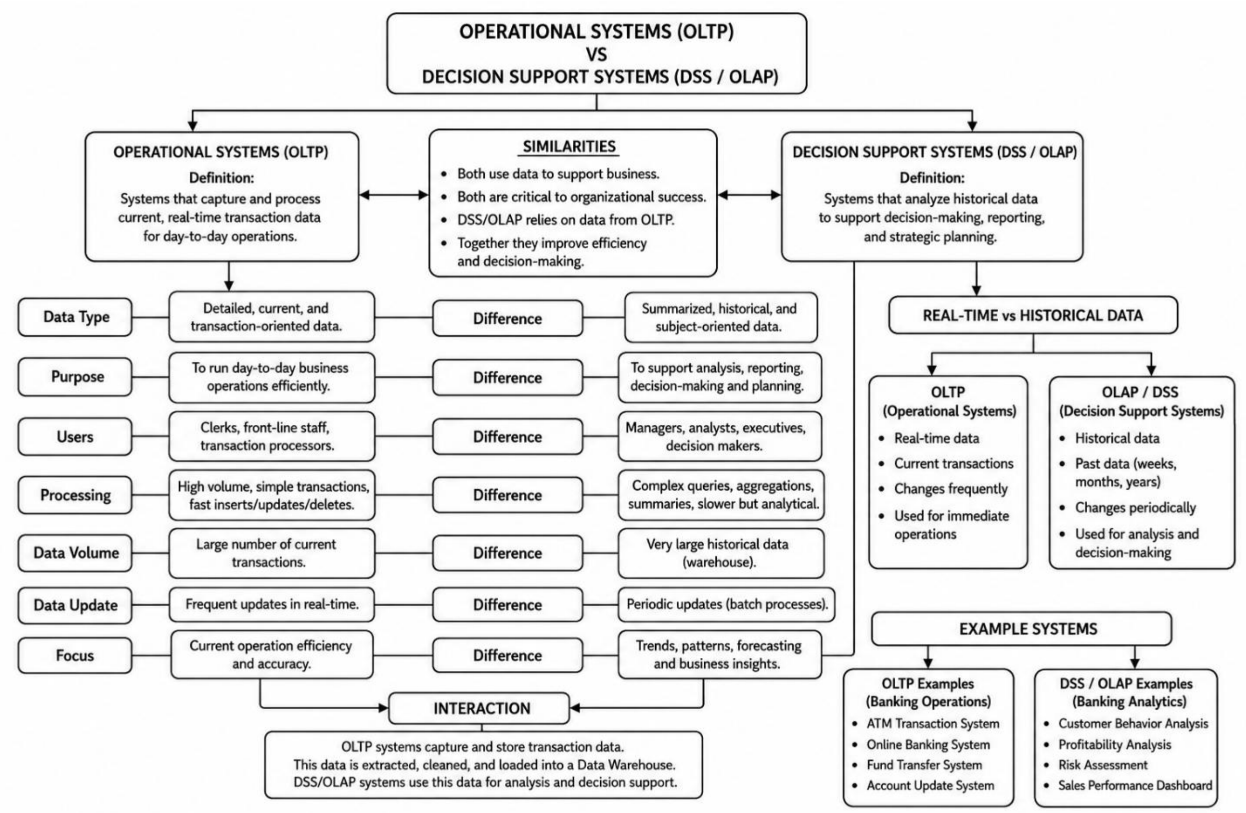
2) Concept Map 2: Operational Systems vs DSS

Task Description:

Develop a comparative concept map that clearly distinguishes Operational Systems (OLTP) and Decision Support Systems (OLAP).

What your map must show:

1. Definitions of OLTP and DSS
2. Differences (data type, purpose, users, processing)
3. Similarities and interactions
4. Real-time vs historical data
5. Example: systems (banking vs analytics)



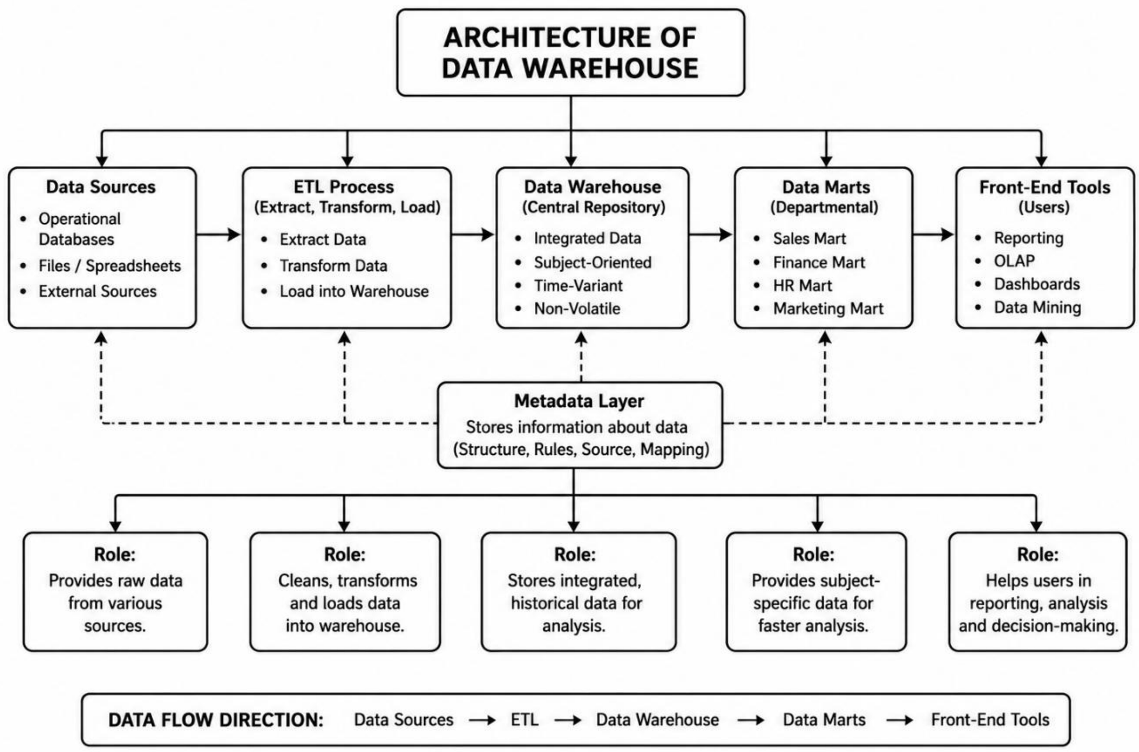
3) Concept Map 3: Architecture of Data Warehouse

Task Description:

Create a hierarchical concept map illustrating the architecture of a Data Warehouse.

What your map must show:

1. Data Sources → ETL → Data Warehouse → Data Marts
2. Metadata layer
3. Front-end tools (reporting, OLAP)
4. Data flow direction
5. Role of each layer



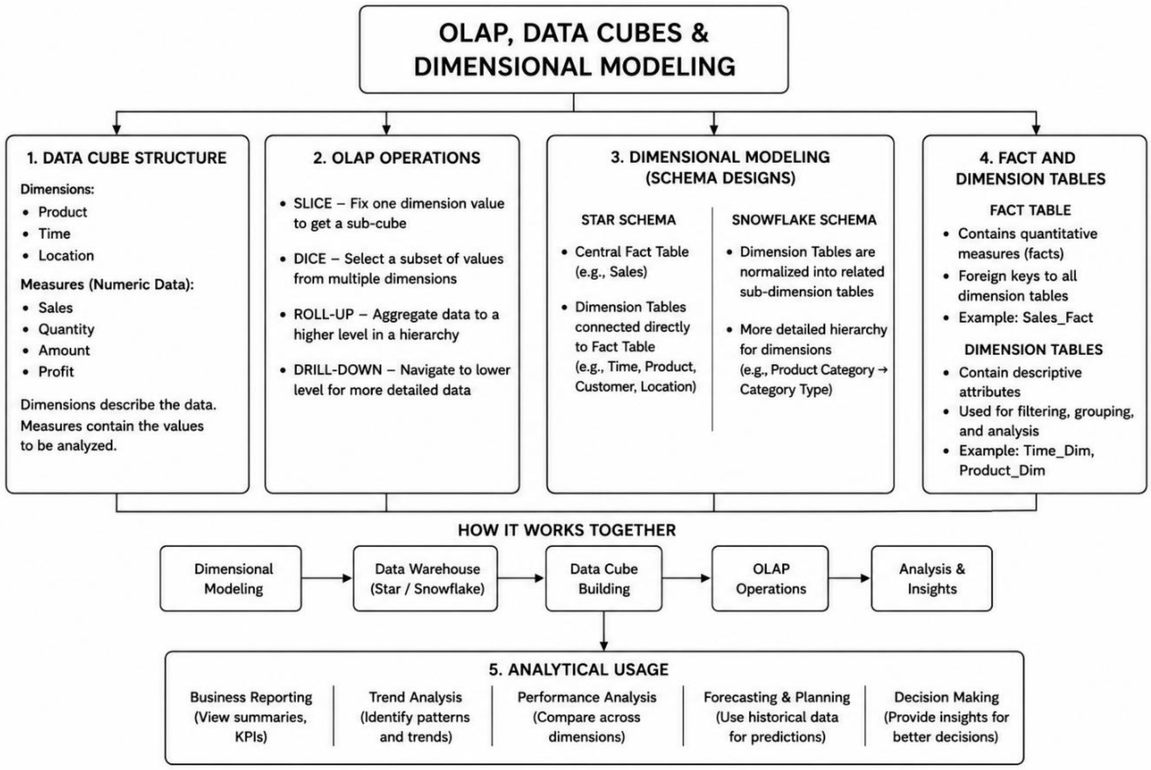
4) Concept Map 4: OLAP, Data Cubes & Dimensional Modeling

Task Description:

Design a concept map that connects OLAP operations, Data Cubes, and Dimensional Modeling techniques.

What your map must show:

1. Data cube structure (dimensions, measures)
2. OLAP operations (slice, dice, roll-up, drill-down)
3. Star schema and snowflake schema
4. Fact and dimension tables
5. Analytical usage

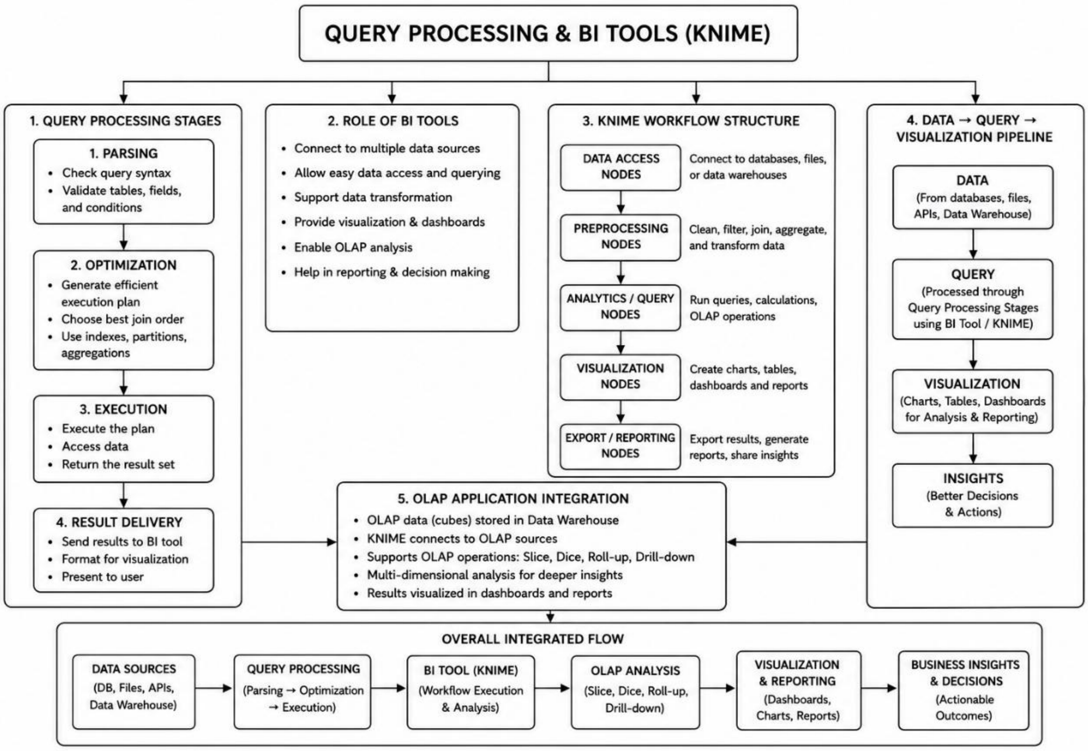


5) Concept Map 5: Query Processing & BI Tools (KNIME)

Task Description:

Construct a concept map
explaining how query processing integrates with BI tools, focusing on KNIME for reporting and OLAP applications.

What your map must show:
1. Query processing stages (Parsing, Optimization, Execution)
2. Role of BI tools
3. KNIME workflow structure
4. Data → Query → Visualization pipeline
5. OLAP application integration


QUERY PROCESSING & BI TOOLS (KNIME)
1. QUERY PROCESSING STAGES
1. PARSING
• Check query syntax
• Validate tables, fields, and conditions
2. OPTIMIZATION
• Generate efficient execution plan
• Choose best join order
• Use indexes, partitions, aggregations
3. EXECUTION
• Execute the plan
• Access data
• Return the result set
4. RESULT DELIVERY
• Send results to BI tool
• Format for visualization
• Present to user
2. ROLE OF BI TOOLS
• Connect to multiple data sources
• Allow easy data access and querying
• Support data transformation
• Provide visualization & dashboards
• Enable OLAP analysis
• Help in reporting & decision making
3. KNIME WORKFLOW STRUCTURE
DATA ACCESS NODES
Connect to databases, files, or data warehouses
PREPROCESSING NODES
Clean, filter, join, aggregate, and transform data
ANALYTICS / QUERY NODES
Run queries, calculations, OLAP operations
VISUALIZATION NODES
Create charts, tables, dashboards and reports
EXPORT / REPORTING NODES
Export results, generate reports, share insights
4. DATA → QUERY → VISUALIZATION PIPELINE
DATA
(From databases, files, APIs, Data Warehouse)
QUERY
(Processed through Query Processing Stages using BI Tool / KNIME)
VISUALIZATION
(Charts, Tables, Dashboards for Analysis & Reporting)
INSIGHTS
(Better Decisions & Actions)
5. OLAP APPLICATION INTEGRATION
• OLAP data (cubes) stored in Data Warehouse
• KNIME connects to OLAP sources
• Supports OLAP operations: Slice, Dice, Roll-up, Drill-down
• Multi-dimensional analysis for deeper insights
• Results visualized in dashboards and reports
OVERALL INTEGRATED FLOW
DATA SOURCES
(DB, Files, APIs, Data Warehouse)
QUERY PROCESSING
(Parsing → Optimization → Execution)
BI TOOL (KNIME)
(Workflow Execution & Analysis)
OLAP ANALYSIS
(Slice, Dice, Roll-up, Drill-down)
VISUALIZATION & REPORTING
(Dashboards, Charts, Reports)
BUSINESS INSIGHTS & DECISIONS
(Actionable Outcomes)